Nicholas Spyrison, Ph.D.

Statistician | Data Scientist

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Career Summary

Statistician & Data Scientist with a background in complex physical and clinical systems. Proven track record in developing enterprise-level R packages and deploying scalable machine learning solutions. Expert in statistical testing and visualization with a focus on optimizing code performance and leading technical teams to accelerate project delivery.

Employment History

03/2023 – Present **Biostatistcian**, *IFF – International Flavors and Fragrances*, Madison, WI.

- Optimized production parameters from Machine Learning models fit to wide production data leading to $440K USD hard savings and $2M USD in soft savings
- Automated colony plate counting with image analysis and predictions from machine learning models, improved accuracy while decreasing technician time
- Produced, maintained, and interpreted 480 exponential decay models of probiotic stability, informing pricing at a plant averaging $460 Million USD annual sales
- Interpretable Machine Learning, statistical testing & consultation, Shiny development, data visualization, and high-throughput reporting for >300 employees on-site beside other sites

06/2022 – 01/2023 **Research Data Scientist**, *University of Queensland*, Clayton, VIC, Australia.

- Published analysis and figures on global kidney health outcomes from CDISC SDTM data structure
- Worked agilely with SMEs, iterating by creating models predicting kidney health

07/2018 – 10/2022 **Research Officer**, *Monash University*, Clayton, VIC, Australia.

- *Data Fluency Instructor*: Lead workshops (10-30 people) on statistical modeling, R, Python, PowerBI, and Git
- *Teaching Assistant*: Post-grad tutorials on high-dimensional data analysis, post-grad data visualization, and undergrad economic statistics
- *Research Officer*: Extract, transform, load, and analyze census and text data

12/2016 – 1/2018 **Business Intelligence Developer**, *CPI Card Group*, Denver, CO.

- SQL reporting; across 4 reporting platforms and 4 servers
- Mentor junior teammates and improve query performance
- Triage and transfer 30+ legacy reports to a modern reporting solution
- Developed supply chain report tracking a cost reduction of more than $4 Million USD

07/2015 – 10/2016 **Business Intelligence Developer**, *UDR Real Estate Investment Trust*, Denver, CO.

- Transition enterprise reporting from WebFOCUS to SQL reports (SSRS) – 10+ reports

11/2014 – 04/2015 **Business Intelligence Analyst**, *Heartland Financial USA*, Dubuque, IA.

- Financial reporting for internal and external audiences

02/2013 – 05/2014 **Delivery Analyst**, *IBM – International Business Machines*, Dubuque, IA.

- Automated ETL with VBA macros for about 0.5 FTE savings across the team
- KPI reporting to teams and management

01/2010 – 08/2012 **Laboratory Technician**, *The Ames Laboratory & Iowa State University*, Ames, IA.

Authored Software (R Packages)

09/2022 **cheem** – Novel analysis and interactive visual for exploring ML interpretability *Link*

10/2020 **spinifex** – ggplot2-like API for composing and exporting animated projections *Link*

Education

02/2018 – 06/2022 **Ph.D. Information Technology & Statistics**, *Monash University*, Clayton, VIC,
Dimension reduction, interactive and animated data visualization, machine learning
Thesis: *Interactive and Dynamic Visualization of High-dimensional Data*.

- Novel interoperability visualization methods for **Fit random forest models on several contemporary datasets** also shown to generalize to other models with **SVM and XGboost**
- Utilized **SHAP and LIME local explanations** to explore local linear variable importance to understand large residuals in the models
- Created **novel interactive animation** facilitating this exploration in a web app
- **Authored R packages** on CRAN as free and open source work to improve reproducibility and contribute back to the community

08/2008 – 05/2012 **B.Sc. Statistics**, *Iowa State University*, minors in Physics and Mathematics, Ames, IA.

Technical Skills

Software R, SQL, Python, Git, GitHub, LaTeX, Excel, VBA

Packages tidyverse, tidymodels, ggplot2, plotly, caret, keras, pandas, scikit-learn

Reporting R Shiny, R Markdown/Quarto, SSRS, Tableau, PowerBI

Awards and Honors

03/2024 IFF Internal Recognition Award – Tier 1
- Interactive modeling, visualization, and export of stability data

06/2022 Invited article for INFORM's OR/MS Tomorrow *Link*

04/2022 RStudio: March "Top 40" New CRAN Packages *Link*

05/2022 Monash Postgraduate Publication Award

10/2020 Melbourne Datathon 2020 – **1st place of several hundred entries** *Insights category* *Link*
- *Changes to Victorian Intraday Electricity Demand Following COVID-19 Restrictions*

11/2018 ACEMS Impact and Engagement Award

07/2018 UseR2018 Datathon – 3rd place
- Spatio-temporal animated visualization of over 7 million avian sightings

2018 – 2022 Faculty Graduate Research Scholarship

2018 – 2022 Co-funded Monash Graduate Scholarship

05/2012 Nominated for Student Employee of the Year

05/2011 Nominated for Student Employee of the Year

05/2008 American Legion Award – High-honor recognition of character

Publications

Doctoral Thesis

[1] Nicholas Spyrison. "Interactive and dynamic visualization of high-dimensional data". en. PhD thesis. Monash University, Mar. 2022. URL: `https://github.com/nspyrison/thesis_ns/blob/master/docs/thesis_ns.pdf`.

Journal Articles

[2] Danyang Dai et al. "Omicron surge impact on acute kidney injury in ICU patients: A study using the ISARIC COVID-19 database". In: *PloS one* 20.11 (2025), e0336843.

[3] Pedro Henrique Franca Gois et al. "AKI in Intensive Care Unit (ICU) Patients in the Omicron Surge: Insights from a Multinational Study: TH-PO1144". In: *Journal of the American Society of Nephrology* 35.10S (2024), pp. 10–1681.

[4] Marina Wainstein et al. "WCN24-1193 VALIDATION OF THE EXTENDED KDIGO DEFINITION TO DIAGNOSE ACUTE KIDNEY INJURY IN A GENERAL HOSPITAL POPULATION USING THE MIMIC-IV DATASET". In: *Kidney International Reports* 9.4 (2024), S261–S262.

[5] Anatoliy Gavrylov et al. "Association of Country Income Level With the Characteristics and Outcomes of Critically Ill Patients Hospitalized With Acute Kidney Injury and COVID-19". In: (2023).

[6] Marina Wainstein et al. "Association of country income level with the characteristics and outcomes of critically ill patients hospitalized with acute kidney injury and COVID-19". In: *Kidney International Reports* (2023).

[7] Nicholas Spyrison, Dianne Cook, and Przemyslaw Biecek. "Exploring local explanations of nonlinear models using animated linear projections". In: *Computational Statistics* 40.2 (2025), pp. 1071–1095.

[8] Nicholas Spyrison, Dianne Cook, and Kim Marriott. "A Study on a User-Controlled Radial Tour for Variable Importance in High-Dimensional Data". In: *arXiv preprint, submitted to TVCG* (Dec. 2022). arXiv:2301.00077 [stat]. DOI: 10.48550/arXiv.2301.00077. URL: http://arxiv.org/abs/2301.00077 (visited on 01/06/2023).

[9] Stuart Lee et al. "The state-of-the-art on tours for dynamic visualization of high-dimensional data". en. In: *WIREs Computational Statistics* (Dec. 2021), p. 21. ISSN: 1939-0068. DOI: 10.1002/wics.1573. URL: https://onlinelibrary.wiley.com/doi/pdf/10.1002/wics.1573 (visited on 12/10/2021).

[10] Nicholas Spyrison and Dianne Cook. "spinifex: an R Package for Creating a Manual Tour of Low-dimensional Projections of Multivariate Data". en. In: *The R Journal* 12.1 (2020), p. 243. ISSN: 2073-4859. DOI: 10.32614/RJ-2020-027. URL: https://journal.r-project.org/archive/2020/RJ-2020-027/index.html (visited on 10/16/2020).

Software (R Packages)

[11] Nicholas Spyrison. *cheem: Interactively Explore the Support of Local Explanations of a Model*. 2022. URL: https://CRAN.R-project.org/package=cheem.

[12] Nicholas Spyrison and Dianne Cook. *spinifex: Manual Tours, Manual Control of Dynamic Projections of Numeric Multivariate Data*. 2021. URL: https://CRAN.R-project.org/package=spinifex.

Conference Proceedings

[13] Nicholas Spyrison. "Animated Linear Projections". In: *Summer 2022*. INFORMS, May 2022.

[14] Dianne Cook, Stuart Lee, and Nicholas Spyrison. "A Showcase of New Methods for High Dimensional Data Viewing with Linear Projections and Sections". In: *Book of Abstracts*. International Federation of Classification Societies, 2022, p. 10.

[15] Nicholas Spyrison, Benjamin Lee, and Lonni Besançon. ""Is IEEE VIS *that* good?" On key factors in the initial assessment of manuscript and venue quality". In: *IEEE AltVis Workshop*. type: article. July 2021. DOI: 10.31219/osf.io/65wm7. URL: https://osf.io/65wm7/ (visited on 08/21/2021).

[16] Madeleine Barrow, Jieyang Chong, and Nicholas Spyrison. "Changes to Victorian Intraday Electricity Demand Following COVID-19 Restrictions". en. In: *Melbourne Datathon 2020, Insights category*. Oct. 2020, p. 4. URL: https://github.com/nspyrison/melb_datathon2020/blob/master/_paper/paper.pdf (visited on 02/09/2022).

[17] Nicholas Spyrison. "spinifex: visualizing local structure of higher dimensions". en. In: Brisbane, Australia, 2018. URL: https://user2018.r-project.org/poster/ (visited on 07/28/2020).

References available on request.